Integration Project

Qube Servo Pendulum

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List of Symbols

Symbol	Description	Unit
Geometric & inertial parameters		
L_0	Length of the first (rotary) arm	m
L_1	Length of the second (pendulum) arm	m
r	Length ratio L_0/L_1	–
m_0	Mass of the first arm	kg
m_1	Mass of the pendulum arm	kg
J_0	Second moment of inertia of the first arm about the θ-axis	$\mathrm{kg\,m^2}$
Generalised coordinates		
θ	Rotation of the first arm (about vertical z-axis)	rad
α	Rotation of the pendulum arm (about its own horizontal axis)	rad
$\dot{\theta},\ \dot{\alpha}$	Angular velocities	$\mathrm{rad\,s^{-1}}$
$\ddot{\theta},\ \ddot{\alpha}$	Angular accelerations	$\mathrm{rad\,s^{-2}}$
Kinematics of a slice s along the pendulum		
s	Distance from the joint along the pendulum shaft	m
P_s	Position vector of the slice	m
v_s	Linear velocity of the slice	$\mathrm{m\,s^{-1}}$
Energy terms		
T_0	Kinetic energy of the first arm	J
T_1	Kinetic energy of the pendulum arm	J
T_s	Kinetic energy density of slice s	$\mathrm{J\,m^{-1}}$
V_0	Potential energy of the first arm (chosen 0)	J
V_1	Potential energy of the pendulum arm	J
L	Lagrangian $T_0 + T_1 - V_0 - V_1$	J
Damping and motor constants		
d_0	Viscous damping about the θ-axis	$\mathrm{N\,m\,s\,rad^{-1}}$
d_1	Viscous damping about the α-axis	$\mathrm{N\,m\,s\,rad^{-1}}$
K_t	Motor torque constant	$\mathrm{N\,m\,A^{-1}}$
K_e	Back-EMF constant	$\mathrm{V\,s\,rad^{-1}}$
R	Motor armature resistance	Ω
V	Applied motor voltage (control input)	V
Matrices used in the linear model		
$A,\ B,\ C,\ D$	Continuous-time state-space matrices	–
A_d, B_d, C_d, D_d	Discrete-time (sampled) state-space matrices	–
M	Symmetric matrix appearing in T_1 (inertia-like)	–
$\bar{M}$	M augmented by k_0 (see Sec. 2)	–
$Q,\ R$	State and input weighting matrices (LQR/MPC)	–
P	Terminal weight (solution of DARE)	–
K	State-feedback gain (LQR)	–
L	Observer gain (Luenberger)	–
H	Hessian of condensed MPC QP	–
K_{lin}	Linear term multiplier in MPC cost	–
Other symbols		
g	Acceleration due to gravity	$\mathrm{m\,s^{-2}}$
h	Sampling time (discretisation interval)	s
N	Prediction horizon (MPC)	–
$u_{\max}$	Magnitude bound on motor voltage	V

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Symbol	Description	Unit
$c_{\max}$	Ellipsoidal terminal-set scalar ($x^\top P x \leq c_{\max}$)	–
$\mathbb{U}$, $\mathbb{X}$	Input and state constraint sets	–
$\mathbb{X}_f$	Terminal invariant set	–
V_f	Terminal cost / Lyapunov function	–
e	Integral tracking error (augmented state)	rad

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Modelling and Identification

This report details the process to model, identify and design two controllers for the inverted cube pendulum produced by Quanser. The pendulum consists of a motor that rotates around a vertical axis. There is a one link connected at its one end to the motor, therefore rotating around the vertical axis. The first link is also free to rotate around its longitudinal axis. At the other end of the first link is connected the second link. It rotates together with the first link around a horizontal, longitudinal axis of the first link. There is one encoder, measuring the angle of rotation of the motor and the second one measuring the angle of rotation around the axis of rotation of the first link.

1.1. Modelling

First, the coordinate system and the definition of the terms are visualized on Figure 1.1. The first rotatory arm has properties indexed $_0$, and the pendulum has properties indexed $_1$. The length of the arms are respectively: L_0 and L_1, the mass of them are: m_0 and m_1, the second moment of inertia of the rotatory arm along the axis of the rotation θ is J_0. The rotation of the rotatory arm is described by the angle θ and the rotation of the pendulum is described by the angle α. The gravity acts in the $-\hat{z}$ direction. The damping caused by the friction due to rotation of θ is d_0 and the due to the rotation caused by a α is d_1. The slice of the cylinder of the pendulum that is defined with distance s from the joint of the two arms is denoted by the index $_s$. It has a mean position of P_s, velocity v_s and kinetic energy T_s. The whole pendulum has kinetic energy T_1 and potential energy V_1, while the rotatory arm has has kinetic energy T_0 and potential energy V_0.

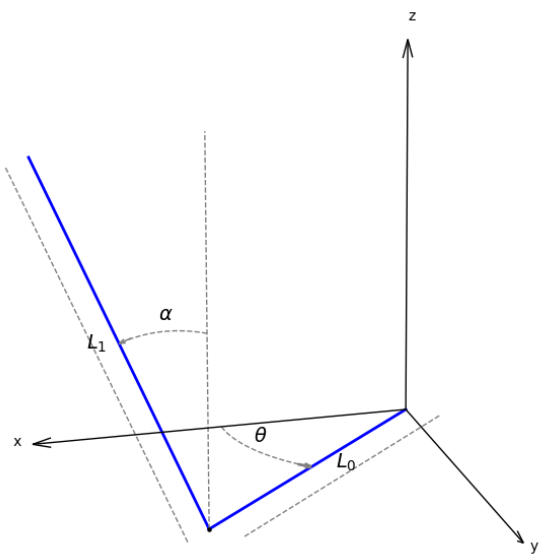

Figure 1.1: Sketch of rotating pendulum system

The modelling of the system takes the approach of least action. The Lagrangian needs to be ex-

pressed as the function of the angles θ and α. The Lagrangian is given by the difference between the kinetic and potential energy:

$$L = T_0 + T_1 - V_0 - V_1 \tag{1.1}$$

Where the terms are expressed as follows with $V_0 = 0$:

$$V_1 = \frac{1}{2} L_1 m_1 g cos(\alpha) \tag{1.2}$$

$$T_0 = \frac{1}{2} \dot{\theta}^2 J_0 \tag{1.3}$$

$$T_1 = \frac{1}{2} \frac{m_1}{L_1} \int_0^{L_1} v_s^T v_s ds \tag{1.4}$$

It is assumed that all significant mass of both rods re distributed in one perfect line segment that has one of its end points at the intersection of the two axis of rotation. The velocity is further expressed as:

$$v_S = \frac{dP_s}{dt} = \frac{\partial P_s}{\partial \theta} \dot{\theta} + \frac{\partial P_s}{\partial \alpha} \dot{\alpha} \tag{1.5}$$

where the position of point P_s is dependent on theta and alpha

$$P_s = L_0 \begin{pmatrix} cos(\theta) \\ sin(\theta) \\ 0 \end{pmatrix} + s \begin{pmatrix} sin(\alpha) sin(\theta) \\ -sin(\alpha) cos(\theta) \\ cos(\alpha) \end{pmatrix} \tag{1.6}$$

This results in the velocity of a point:

$$v_S = (L_0 C_0 + s C_1) \dot{\theta} + s C_2 \dot{\alpha} \tag{1.7}$$

with C_0, C_1 and C_2 being defined as:

$$\begin{pmatrix} C_0 & C_1 & C_2 \end{pmatrix} = \begin{pmatrix} -sin(\theta) & sin(\alpha) cos(\theta) & cos(\alpha) sin\theta) \\ cos(\theta) & sin(\alpha) sin(\theta) & -cos(\alpha) cos(\theta) \\ 0 & 0 & -sin(\alpha) \end{pmatrix} \tag{1.8}$$

Which leads to the expression of the kinetic energy of the slice:

$$\begin{aligned} \frac{dT_s}{ds} \cdot \frac{L_1}{m_1} 2 = {} & \dot{\theta}^2 L_0^2 C_0^T C_0 \\ & + s \left(L_1 C_0^T \dot{\theta} (C_1 \dot{\theta} + C_2 \dot{\alpha}) + (C_1^T \dot{\theta} + C_2^T \dot{\alpha}) C_0 L_1 \dot{\theta} \right) \\ & + s^2 \left((C_1^T \dot{\theta} + C_2^T \dot{\alpha})(C_1 \dot{\theta} + C_2 \dot{\alpha}) \right) \end{aligned}$$

In order to get the kinetic energy of the entire pendulum, this is integrated to gain:

$$T_1 = \frac{1}{2} \frac{m_1}{L_1} \int_0^{L_1} \frac{dT_s}{ds} \frac{L_1}{m_1} 2 ds \tag{1.9}$$

which is equal to

$$\begin{aligned} T_1 = \frac{1}{2} \frac{m_1}{L_1} \Bigg(& L_1 \left[\dot{\theta}^2 L_0^2 C_0^T C_0 \right] \\ & + \frac{1}{2} L_1^2 \left[L_1 C_0^T \dot{\theta} (C_1 \dot{\theta} + C_2 \dot{\alpha}) + (C_1^T \dot{\theta} + C_2^T \dot{\alpha}) C_0 L_1 \dot{\theta} \right] \\ & + \frac{1}{3} L_1^3 \left[(C_1^T \dot{\theta} + C_2^T \dot{\alpha})(C_1 \dot{\theta} + C_2 \dot{\alpha}) \right] \Bigg) \end{aligned}$$

Which can be simplified into a quadratic expression of the angular velocities as

$$T_1 = \frac{1}{2} m_1 L_1^2 \begin{pmatrix} \dot{\theta} & \dot{\alpha} \end{pmatrix} M \begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} \tag{1.10}$$

with $M \epsilon S^n$ is a symmetric matrix and $r = L_0/L_1$

$$M := \begin{pmatrix} r^2 C_0^T C_0 + r C_0^T C_1 + \frac{1}{3} C_1^T C_1 & \frac{1}{2} r C_0^T C_2 + \frac{1}{3} C_1^T C_2 \\ \frac{1}{2} r C_0^T C_2 + \frac{1}{3} C_1^T C_2 & \frac{1}{3} C_2^T C_2 \end{pmatrix} \tag{1.11}$$

where the elements of M are distinguished for later use as:

$$M = \begin{pmatrix} M_{11} & M_{12} \\ M_{12} & M_{22} \end{pmatrix} \tag{1.12}$$

Since later in the derivation the partial derivatives of M with respect to θ and α are required they are also expressed. The derivative of M with respect to θ is 0, which aligns with intuition as if the same motion takes place but shifted horizontally, in θ, it doesn't change the dynamics of the system. The derivative with respect to α on the other hand has the form

$$\frac{\partial M}{\partial \alpha} = \begin{pmatrix} M_1 & M_2 & M_3 \end{pmatrix} \left(\frac{\partial C_0^T}{\partial \alpha} \begin{pmatrix} C_0 \\ C_1 \\ C_2 \end{pmatrix} + C_0^T \begin{pmatrix} \frac{\partial C_0}{\partial \alpha} \\ \frac{\partial C_1}{\partial \alpha} \\ \frac{\partial C_2}{\partial \alpha} \end{pmatrix} \right) \tag{1.13}$$

$$+ \begin{pmatrix} M_4 & M_5 \end{pmatrix} \left(\frac{\partial C_1^T}{\partial \alpha} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} + C_1^T \begin{pmatrix} \frac{\partial C_1}{\partial \alpha} \\ \frac{\partial C_2}{\partial \alpha} \end{pmatrix} \right) \tag{1.14}$$

$$+ M_6 \cdot 2 \cdot \frac{\partial C_2^T}{\partial \alpha} C_2 \tag{1.15}$$

with

$$M_1 = \begin{pmatrix} r^2 & 0 \\ 0 & 0 \end{pmatrix}; M_2 = \begin{pmatrix} r & 0 \\ 0 & 0 \end{pmatrix}; M_3 = \begin{pmatrix} 0 & \frac{1}{2} r \\ \frac{1}{2} r & 0 \end{pmatrix}; M_4 = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & 0 \end{pmatrix}; M_5 = \begin{pmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & 0 \end{pmatrix}; M_6 = \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{3} \end{pmatrix} \tag{1.16}$$

At this point everything is defined to express the Lagrangian:

$$L = \frac{1}{2} \dot{\theta}^2 J_0 + \frac{1}{2} m_1 L_1^2 \begin{pmatrix} \dot{\theta} & \dot{\alpha} \end{pmatrix} M \begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} - \frac{1}{2} L_1 m_1 g cos(\alpha) \tag{1.17}$$

In order to find the dynamics using the principle of least action, the derivative of the Lagrangian with respect to the free variables θ and α and $\dot{\theta}$ and $\dot{\alpha}$ are defined as:

$$\frac{\partial L}{\partial \theta} = 0 \tag{1.18}$$

$$\frac{\partial L}{\partial \alpha} = \frac{1}{2} L_1 m_1 g sin(\alpha) + \frac{1}{2} m_1 L_1^2 \begin{pmatrix} \dot{\theta} & \dot{\alpha} \end{pmatrix} \frac{\partial M}{\partial \alpha} \begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} \tag{1.19}$$

$$\frac{\partial L}{\partial \dot{\theta}} = \dot{\theta} J_0 + m_1 L_1^2 \begin{pmatrix} 1 & 0 \end{pmatrix} M \begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} \tag{1.20}$$

$$\frac{\partial L}{\partial \dot{\alpha}} = m_1 L_1^2 \begin{pmatrix} 0 & 1 \end{pmatrix} M \begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} \tag{1.21}$$

The principle of least actions states that:

$$\begin{pmatrix} \tau_\theta \\ \tau_\alpha \end{pmatrix} = \frac{d}{dt} \begin{pmatrix} \frac{\partial L}{\partial \dot{\theta}} \\ \frac{\partial L}{\partial \dot{\alpha}} \end{pmatrix} - \begin{pmatrix} \frac{\partial L}{\partial \theta} \\ \frac{\partial L}{\partial \alpha} \end{pmatrix} \tag{1.22}$$

Where the terms can be expressed in a more simpler form with:

$$\bar{M} = M + \begin{pmatrix} k_0 & 0 \\ 0 & 0 \end{pmatrix} \tag{1.23}$$

with $k_0 = J_0/(L_1^2 m_1)$, which leads to:

$$\frac{1}{L_1^2 m_1}\frac{d}{dt}\begin{pmatrix} \frac{\partial L}{\partial \dot{\theta}} \\ \frac{\partial L}{\partial \dot{\alpha}} \end{pmatrix} = \frac{d}{dt}\left(\bar{M}\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix}\right) = \bar{M}\begin{pmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{pmatrix} + \frac{\partial M}{\partial \alpha}\dot{\alpha}\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} \tag{1.24}$$

Which results in

$$\frac{1}{L_1^2 m_1}\begin{pmatrix} \tau_\theta \\ \tau_\alpha \end{pmatrix} = \bar{M}\begin{pmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{pmatrix} + \frac{\partial M}{\partial \alpha}\dot{\alpha}\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} - \frac{g}{2L_1}\begin{pmatrix} 0 \\ sin(\alpha) \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 0 & 0 \\ \dot{\theta} & \dot{\alpha} \end{pmatrix}\frac{\partial M}{\partial \alpha}\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} \tag{1.25}$$

which is rewritten to express the angular accelerations as:

$$\begin{pmatrix} \ddot{\theta} \\ \ddot{\alpha} \end{pmatrix} = \bar{M}^{-1}\left[\frac{1}{L_1^2 m_1}\begin{pmatrix} \tau_\theta \\ \tau_\alpha \end{pmatrix} + \frac{g}{2L_1}\begin{pmatrix} 0 \\ sin(\alpha) \end{pmatrix} + \frac{1}{2}\begin{pmatrix} -2\dot{\alpha} & 0 \\ \dot{\theta} & -\dot{\alpha} \end{pmatrix}\frac{\partial M}{\partial \alpha}\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix}\right] \tag{1.26}$$

The torque is expressed as the summation of the friction, the cable acting as a spring, and the engine exerting a torque that is linearly dependent on the voltage and the rotation of the engine as:

$$V(t) = L\,\dot{i}(t) + R\,i(t) + K_e\,\dot{\theta} \tag{1.27}$$

$$\tau(t) = K_t i(t) \tag{1.28}$$

As the dynamics of the motor are much faster than that of the pendulum, armature inductance is assumed to go to zero.

$$i \approx \frac{V - K_e\,\dot{\theta}}{R}, \tag{1.29}$$

$$\tau = K_t\,i = \frac{K_t}{R}\left(V - K_e\,\dot{\theta}\right). \tag{1.30}$$

These terms are then expressed in the same order

$$\frac{1}{L_1^2 m_1}\begin{pmatrix} \tau_\theta \\ \tau_\alpha \end{pmatrix} = \begin{pmatrix} -k1 & 0 \\ -k_2 & -k_3 \end{pmatrix}\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} + \begin{pmatrix} -k_4 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} \theta \\ \alpha \end{pmatrix} + \begin{pmatrix} k_5 \\ 0 \end{pmatrix}V \tag{1.31}$$

Since linear models offer simple, linear control strategies such as pole-placement, LQR, MPC, among others, the dynamics is linearized around the unstable point $\theta, \alpha, \dot{\theta}, \dot{\alpha}, V = 0, 0, 0, 0, 18$ and the stable point $\theta, \alpha, \dot{\theta}, \dot{\alpha}, V = 0, \pi, 0, 0, 18$. The linearization for both can be expressed as terms of the constant k values and $\bar{M}$. which is dependent on angles, therefore the difference between the two system is only the different value of $\bar{M}$. The states, and the linear model is defined as:

$$\begin{pmatrix} \ddot{\theta} \\ \ddot{\alpha} \\ \dot{\theta} \\ \dot{\alpha} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ I & 0 \end{pmatrix}\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \\ \theta \\ \alpha \end{pmatrix} + \begin{pmatrix} B_1 \\ 0 \end{pmatrix}V \tag{1.32}$$

with the blocks of A and B being defined as:

$$A_{11} = \bar{M}^{-1}\begin{pmatrix} -k_1 & 0 \\ -k_2 & -k_3 \end{pmatrix} \tag{1.33}$$

$$A_{12} = \bar{M}^{-1}\begin{pmatrix} -k_4 & 0 \\ 0 & k_5 \end{pmatrix} \tag{1.34}$$

$$B_1 = \bar{M}^{-1}\begin{pmatrix} k_6 \\ 0 \end{pmatrix} \tag{1.35}$$

The initial values of the k constants are estimated based on the prior knowledge on the equipment as

$$r = 0.67L_1/L_0 \qquad k_0 = \frac{1}{3}L_2^2 m_2/(L_1^2 m_1) \qquad k_1 = 0.1$$
$$k_2 = 0.2 \qquad k_3 = 0.15 \qquad k_4 = 0.2$$
$$k_5 = 1000 \qquad k_6 = \frac{1}{2}g/L_1$$

resulting in the initial guess of the system defined in the `idgrey` function:

$$A = \begin{pmatrix} -0.1215 & -0.3646 & -12.1527 & 69.3129 \\ -0.1823 & -1.1469 & -18.2291 & 218.0391 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1154.5 \\ 1189.2 \\ 0 \\ 0 \end{pmatrix} \tag{1.36}$$

1.2. Calibration

The signals of the α encoder are offset and scaled. A linear correction is found sufficient to correct the signal values. Setting $\alpha = 0$ as the middle of the scale and two extreme points $\alpha = -135°$ and $\alpha = 135°$. The correction in radians is then:

$$\bar{\alpha} = \frac{3\pi}{4}\alpha \tag{1.37}$$

1.3. Identification

In order to obtain a usable model of the system, its parameters need to be identified using the physical set up. That is done using the Grey-box method.

1.3.1. Choice of the method

- **Physical meaning** Black-box ARX or subspace methods offer excellent numerical fits, but the resulting coefficients bear no mechanical meaning. With a grey-box model the estimated parameters $k_0 - k_6$ and r map directly to moments of inertia, viscous friction and motor constants of the pendulum setup.

- **Efficiency** Because the structural form is fixed, far fewer data points are required to pin down seven scalars than to identify all A, B, C, D entries of a comparable black-box state-space model.

- **Consistency** Mass and energy couplings enforced by the symbolic equations guarantee passivity and the correct θ–α interaction—constraints a pure numerical fit could easily violate.

1.3.2. Method

1. **Parameterized state equations.** Continuous-time dynamics of the form

$$\dot{x} = A(k)\,x + B(k)\,u, \qquad y = C\,x \tag{1.38}$$

   Linearised about the equilibrium point $(\alpha_0, \theta_0) = (\pi, 0)$, were implemented in an `m`-file, with $x = [\dot{\theta}, \dot{\alpha}, \theta, \alpha]^T$ and the 10-element parameter vector $k = [a_{11}, a_{12} \dots, , b_2]^T$.

2. **Grey model wrapper.** The file handle, sampling time h and the initial guess $k^{(0)}$ (Equation 1.36) were wrapped in an `idgrey` object.

3. **Non-linear LSQ fitting.** `greyest` minimises $J(k) = \sum_t \|y(t) - \hat{y}(t,k)\|_2^2$ via Levenberg–Marquardt. Numerical Jacobians are computed by finite differences; step sizes were auto-tuned by the toolbox.

4. **Validation** The converged parameters $k^\star$ (Equation 1.39) were validated on two unseen input sequences (frequency sweep and composite signal).

5. **Unstable equilibrium** The identified model is then adapted to the unstable equilibrium case, with $(\alpha_0, \theta_0) = (0, 0)$, by flipping the signs of parameters $(a_{21}, a_{12}, a_{23}, a_{14})$.

1.3.3. Implementation

In order to identify the system with the initial guess, an input signal needs to be designed. A good input signal results in a model of the system that tracks the input well but also generalizes well to different signals.

First signals available in Matlab are tried such as a square wave or sawtooth. They can offer a good fit but do not generalize well to different input signals. That is because those simple inputs are not persistently exciting and so do not model all the modes of the system well.

In order to ensure the persistent excitement, input signals that are made up of different signals and different frequencies are tried. It is found that the set up does not respond well to signals above the frequencies of 4Hz. It also shouldnt be necessary to go above that in stabilising the system. That is why the input signals are made up of components with frequencies no higher than 4Hz.

Through trial and error, it is found that a signal made up of the following offers sufficient performance.

Signal	Frequency (Hz)	Amplitude (V)
Frequency Sweep	1–4	0.0045
Sine	0.5	0.0015
Square	1.4	0.0015
Square	3.5	0.0015
Sawtooth	2.1	0.0015

Table 1.1: Signal types with corresponding frequencies and amplitudes

Superimposing those signals results in the input voltage amplitude of Figure 1.2 (with 0 equivalent to the nominal voltage around which the system is linearized). The input lasts 60 seconds.

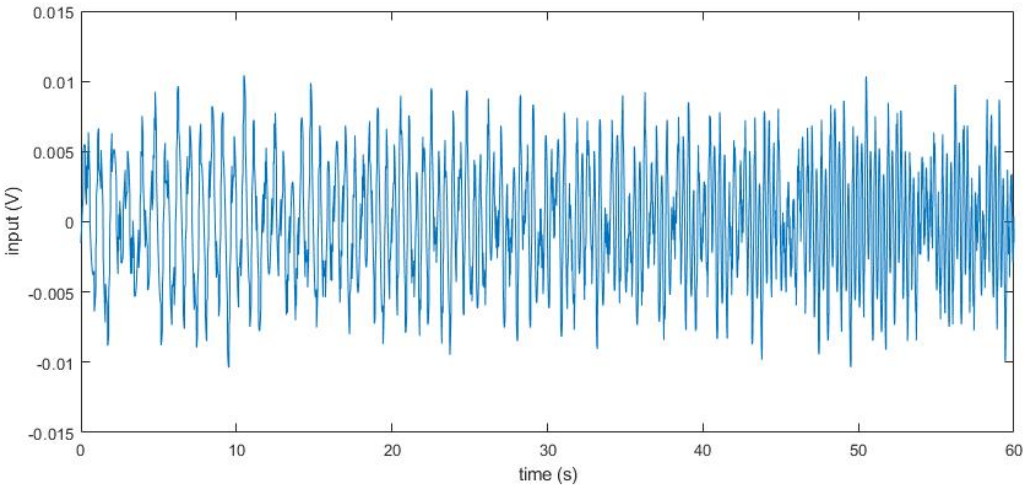

Figure 1.2: Input signal

The resultant identified system is then:

$$A = \begin{pmatrix} -2.8 & -0.6046 & -43.7 & 107 \\ -0.1932 & -1.749 & -46,25 & 231.6 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1133 \\ 1139 \\ 0 \\ 0 \end{pmatrix} \tag{1.39}$$

Which corresponds to the model parameters

Optimal parameters: $k_0 = 0.276139274$, $k_1 = 1.96602143$, $k_2 = -0.873814174$,
$k_3 = 0.301178062$, $k_4 = 17.4249115$, $k_5 = 440.092256$,
$k_6 = 82.6848423$. $r = 0.670210831$

1.3.4. Results

In Figure 1.3, Figure 1.4 the measured data of α and θ is compared with the data simulated with the identified model.

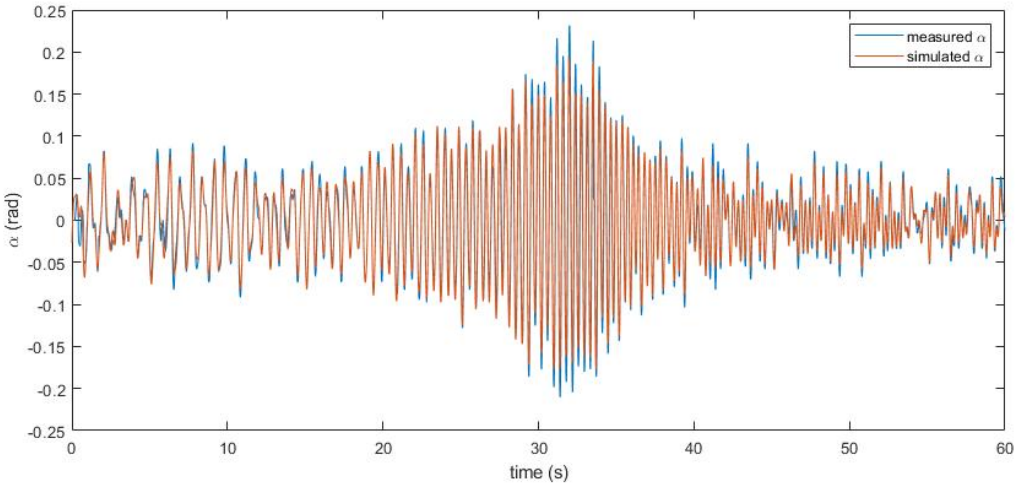

Figure 1.3: Comparing the measured and simulated α data for the identifying input signal

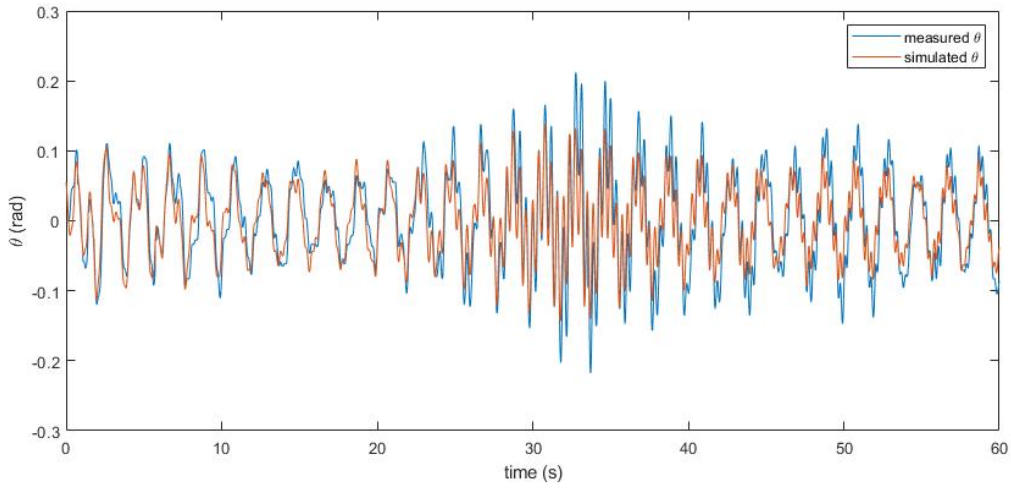

Figure 1.4: Comparing the measured and simulated θ data for the identifying input signal

In order to quantify how well the system is modelled, Variance Accounted For

$$VAF = \left(1 - \frac{var(y - \hat{y})}{var(y)}\right) \cdot 100\% \tag{1.40}$$

and Root Mean Squared Error are calculated.

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}} \tag{1.41}$$

The performance of the identified model on the identification input is thus quantified as:

Output	VAF (%)	RMSE
θ	75.68	0.03418
α	94.15	0.0141

Table 1.2: Identified model performance on the identifying signal

The values of θ are worse than those of α, but should be sufficient given α is more important for stabilizing the system.

1.4. Validation

In order to ensure that the system is correctly identified, the performance of the obtained model is checked for different input signals.

1.4.1. Frequency sweep

Firstly a frequency sweep is checked until higher frequencies:

Signal	Frequency (Hz)	Amplitude (V)
Frequency Sweep	1–8	0.005

Table 1.3: Input signal frequency sweep

Resulting in the following performance parameters:

Output	VAF (%)	RMSE
θ	52.84	0.0161
α	92.06	0.0089

Table 1.4: Identified model performance on the input signal

The performance of the system for this signal is slightly worse than for the identifying signal but decent nonetheless.

In Figure 1.6, Figure 1.7 can also be noted that after the frequency of 4.5Hz is reached, the performance of the system significantly diverges from that of the model. The model wasn't identified with a signal with frequencies with that regime but also it seems that this is a limit, beyond which the system cannot be actuated reliably (as it significantly and unexpectedly diverges).

1.4.2. Designed signal

Secondly, another signal is created made up of different frequencies:

Signal	Frequency (Hz)	Amplitude (V)
Frequency Sweep	1–3	0.002
Sawtooth	0.7	0.002
Sine	1.4	0.002
Square	2.1	0.002
Square	3.5	0.002

Table 1.5: Input signal

Resulting in the following performance parameters:

Output	VAF (%)	RMSE
θ	73.18	0.0217
α	85.56	0.0137

Table 1.6: Identified model performance on the input signal

The performance of the model for this signal is very decent as well. It is therefore concluded that the model generalizes well and so can be used for controller design.

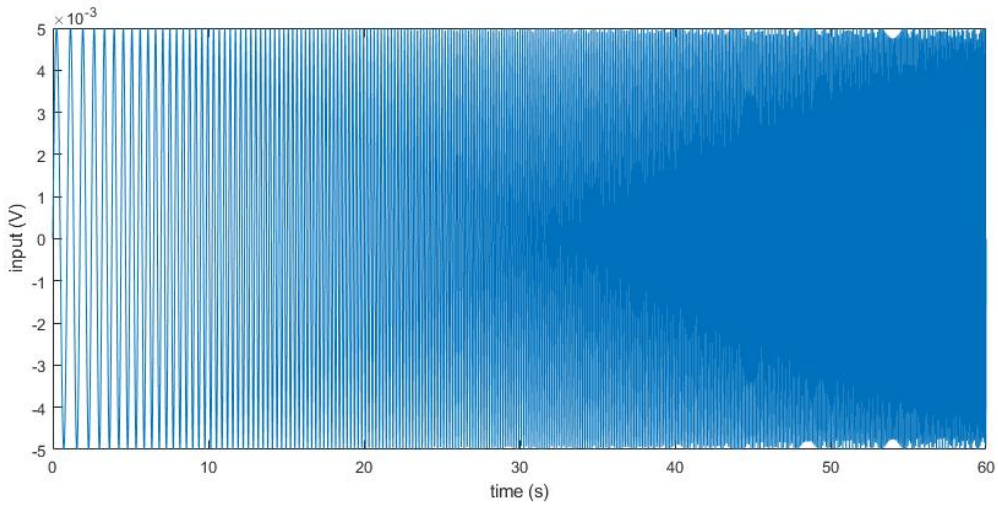

Figure 1.5: Input signal

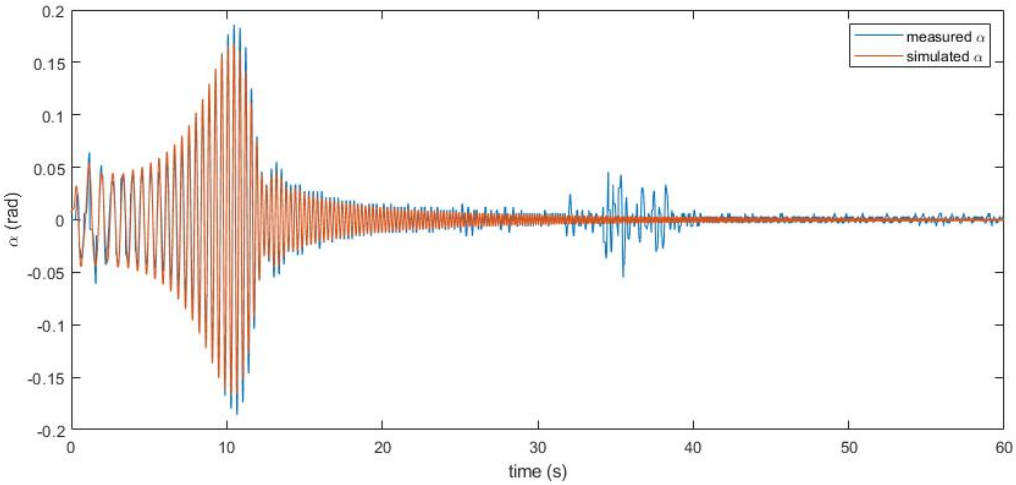

Figure 1.6: Comparing the measured and simulated α data for the input signal

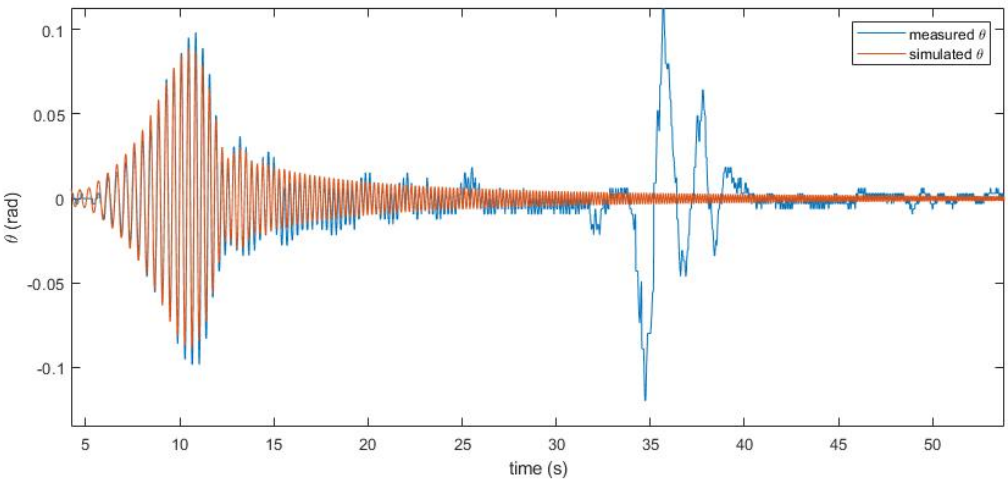

Figure 1.7: Comparing the measured and simulated θ data for the input signal

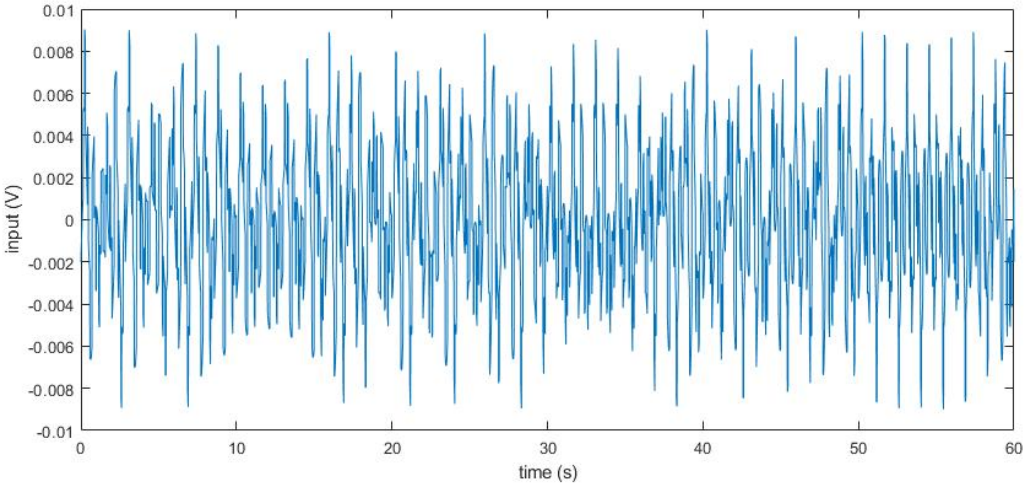

Figure 1.8: Input signal

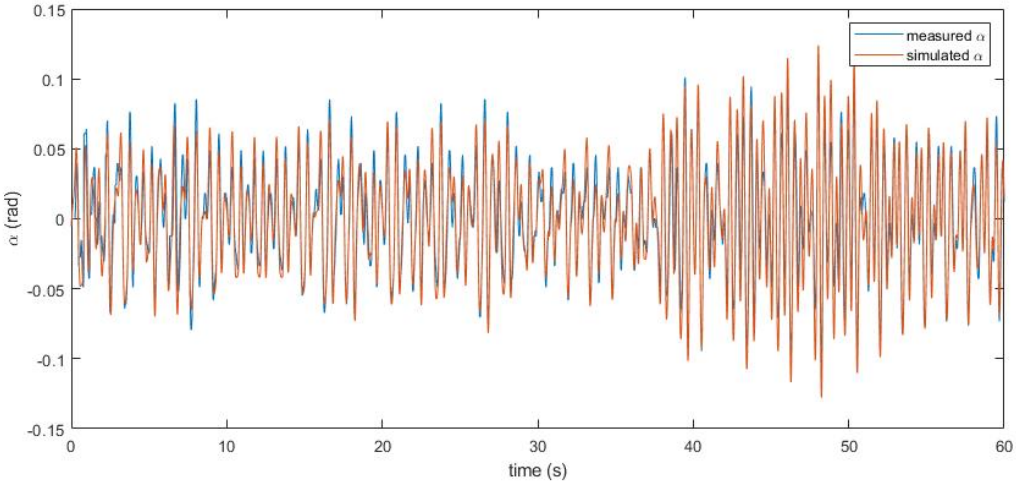

Figure 1.9: Comparing the measured and simulated α data for the input signal

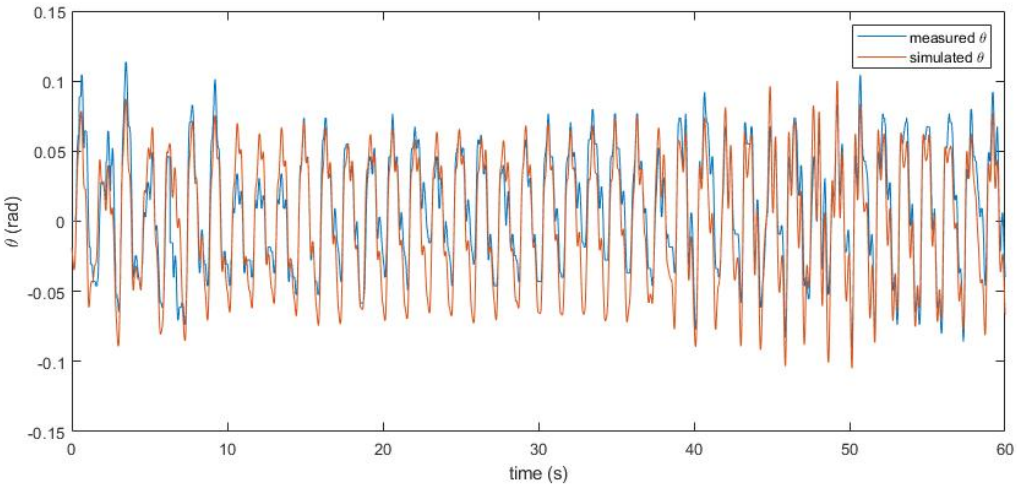

Figure 1.10: Comparing the measured and simulated θ data for the input signal

2

Controller Design

The model is identified, the consequent step is to design controllers, starting with the most simple one, the pole placed state-feedback controller. In order to access the full state information the observer is designed first.

2.1. Observer

Two of the states are available directly via measurements: θ and α, the two angles, which correspond to the 3rd and 4th states: $x(3), x(4)$. The states left to be determined are the angular accelerations $\dot{\theta}$ and $\dot{\alpha}$ which correspond to $x(1)$ and $x(2)$. Since the dynamics is available in this process it is exploited such that the measurements tell us the values of the missing states via the Lundberg observer. While using this observer the following dynamics describes the evolution of the error:

$$e_{k+1} = (A - LC)e_k \tag{2.1}$$

This simple, separated form of the error dynamics allows for a simple design of the observer via the pole placement of the present autonomous system. Since we are designing every controller in discrete time the observer is also designed in discrete time, meaning that the poles of the error dynamics need to be contained in the unit circle in order to drive the error to zero. Another consideration is that the controller will act based on the observed states, meaning that the convergence of error should be quick enough such that it doesn't produce an oscillatory behaviour with the controlled system. Due to this reason the poles of the observer should be around 10 times faster than of the controller. Since the controller is not designed yet the observer poles are determined in with respect to the controllers in the consequent sections.

2.2. Linear Quadratic Regulator

The first controller that is employed at the unstable equilibrium is the Linear Quadratic Regulator (LQR). For LQR the design parameters are the Q and R weight matrices as it solves the optimization problem:

$$u = \arg\min_{u} x^{\top} Q x + u^{\top} R u \tag{2.2}$$

This problem has a well know solution which is based on a linear state-feedback gain for both discrete and continuous systems. For the discrete system the solution is given by $u_\star = K_\star x$

$$P = A^T P A - A^T P B \left(R + B^T P B\right)^{-1} B^T P A + Q \tag{2.3}$$

$$K_\star = -\left(R + B^T P B\right)^{-1} B^T P A \tag{2.4}$$

2.2.1. Implementing an integrator

For reference tracking LQR introduces a persistent offset error which is mitigated by adding an integrator term to the system. For a reference tracking the origin of the coordinate system is shifted by Equation 2.5.

$$x_{new} = x - x_{ref} \tag{2.5}$$

For the LQR systems this introduces a steady state error which is mitigated by adding an integral term to it such as Equation 2.6. This is used for all the reference tracking plots in the following subsection, as it is easier to compare the results if the different LQR behaviours if they produce the same mean. The same set up is used for MPC as well.

$$x_{new}^{(k)} = x^{(k)} - x_{ref} + g_i h \sum_{j=0}^{k-1} x^{(j)} - x_{ref} \tag{2.6}$$

The integrator is implemented on the θ state, before it is fed into the controller. The integrator is of the form:

$$G_i(z) = \frac{g_i T_s}{z-1} \tag{2.7}$$

It is implemented in all the subsequent controller tunings, as it allows for better comparison of the results - different possible offsets do not introduce additional variability in the results.

The gain of the integrator term g_i is tuned for $Q = diag(0, 0, 7, 0), R = 100$. The gain of the integrator influences how aggressive the accumulated error term applied is. Figure 2.1 shows that the optimal value lies within 0 and 1 and for the tested options $g_i^\star = 0.3$ is the optimal one.

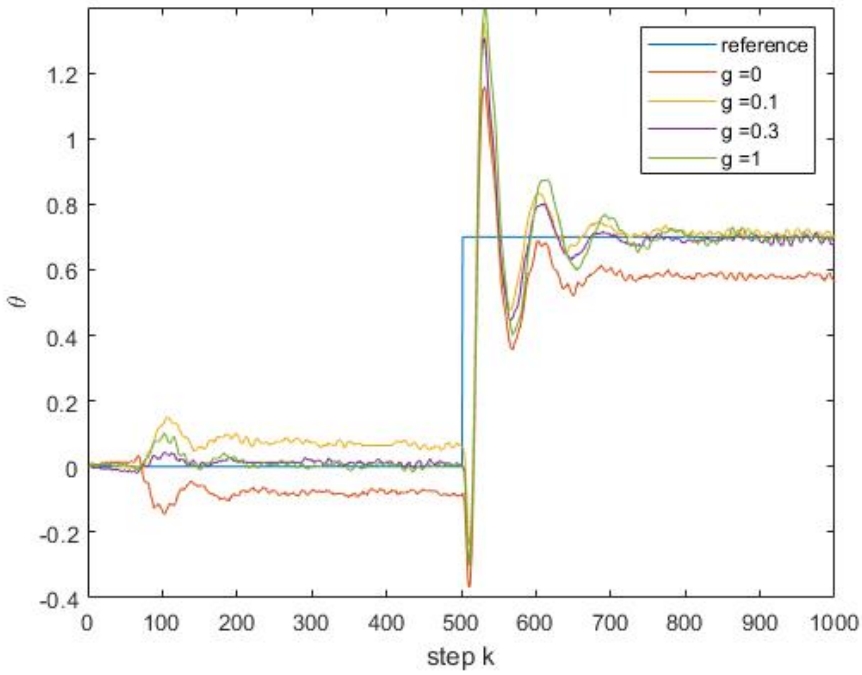

Figure 2.1: Gain tuning of the Integrator term for LQR

2.2.2. Tuning LQR

Theoretically any choice of Q and R reduces a stabilizing result; however, since the model includes imperfections the weight matrices have to be tuned in such a way that the system is stabilized and that the performance is optimized. First, the strategy was to focus on α, as that is the unstable state. Leading to a starting Q and R value of Equation 2.8. Since Q is a 4×4 matrix, even though it is symmetric is still has 10 independent entries, meaning that the tuning is can be very difficult. Due to the time constraints of the project we were focusing only on tuning the diagonal entries of Q such that all the different configurations were examined.

$$Q = diag(0.001, 0.001, 0.1, 2) \quad R = 10 \tag{2.8}$$

According to this strategy the effect of the following changes were examined: changing R, changing the diagonal elements of Q corresponding to the angular velocities, changing the values of Q corresponding to both α and β, scaling everything by the scalar. As expected from theoretical results scaling didn't have an effect on the result. Putting non negligible weights on the values for the angular velocities

resulted in very high frequency behaviour, which is not captured well in our model to neglecting the term in the torque that is induced by high frequencies of θ. Thus it became clear that the first two diagonal elements of Q should either be 0 or a very small number.

First various LQR controllers for high weights on α and low weights on θ and with a low R are produced and their tracking of a step signal are displayed at Figure 2.2. The figure shows that having a zero weight for θ has a negative effect on the performance, which can be mitigated by a high weight for α; however, this introduces high frequency oscillations. To mitigate the issue the value of R is increased in the next batch.

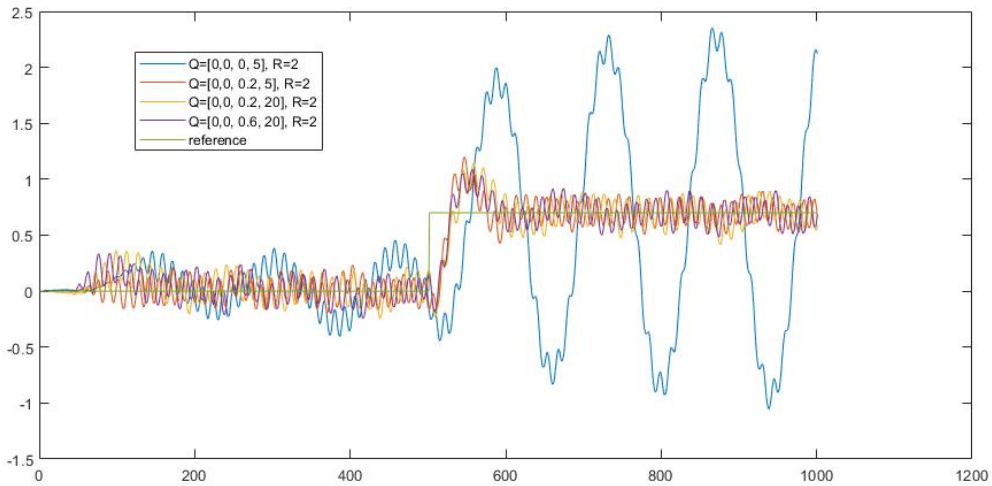

Figure 2.2: LQR with high weights on θ and α and low on u

Figure 2.3 shows the tuning process for high α weights with different R values. The effect of R was well behaving. Under higher R values the motion was smoother, less oscillations at high frequency, but at the cost of an increased settling time. By tuning for R and the wight on α it is clear that the solution lies in another approach, as no combination of these two produces the desired results. Introducing a weight for θ mitigates the high frequency oscillation while it keeps the amplitude low, therefore in the next batch it is tuned for.

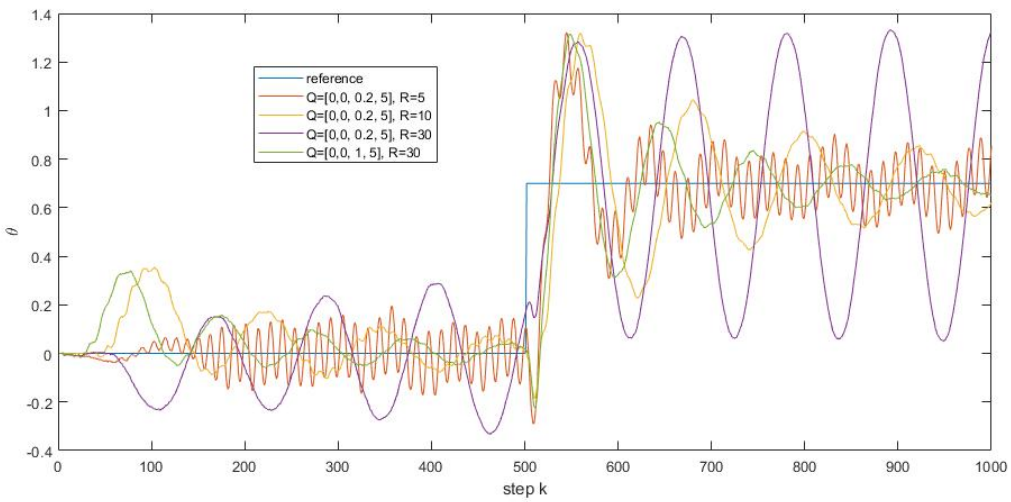

Figure 2.3: LQR with high weights on α and u

As a new approach the aim is stabilize α only through the stabilization of θ. Figure 2.4 shows that this approach minimizes the presence of high frequency oscillations, while tuning for R to be relatively higher introduces low frequency oscillations. This issue is mitigated at $\theta_w = 6, R = 100$. Increasing θ_w

above 8 reintroduces the high frequency oscillations, therefore for an R value of 100 the ideal value for θ_w is $\in [6, 8]$.

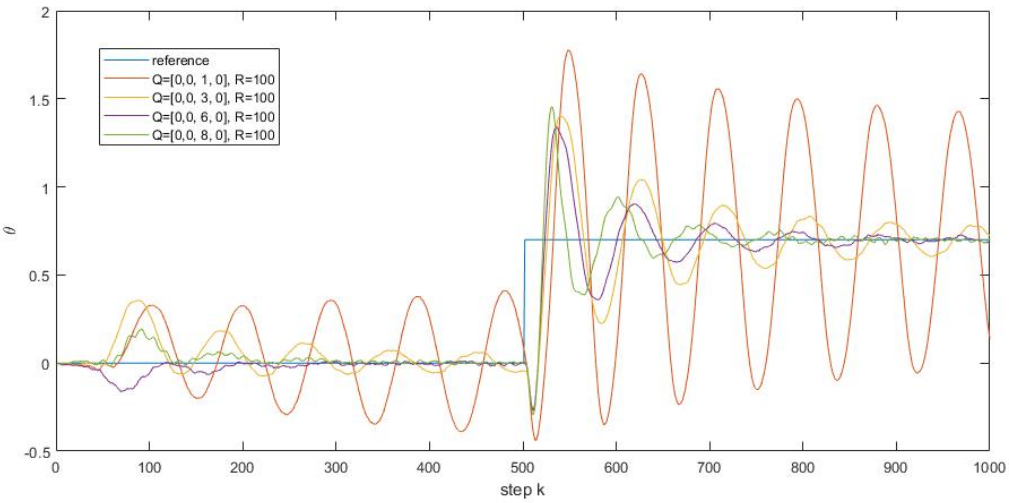

Figure 2.4: LQR with weights on only θ and u

2.2.3. Control Architecture

So far two types of well behaving controllers were gained from the tuning process. One performs well at keeping the states close to the origin with minimal oscillations, but is sensitive to disturbances, while the other one is more robust but introduces high amplitude oscillations with a high settling time. In order to use the advantages of both they can be combined by a control architecture. For α values that deviate only a little bit from the origin the well stabilizing LQR is used, for larger α values the robust one is used, and for α that exceed 1 radian the system is turned off for safety reasonsTable 2.1. For this process two LQR-s are tuned to Equation 2.9 and produces the disturbance rejection of Figure 2.5. In this plot the disturbance is applied manually and it shows that the dual set up has a higher performance.

Regime	Control Strategy
$\lvert\alpha\rvert \leq 0.1$	LQR_1
$0.1 < \lvert\alpha\rvert \leq 1$	LQR_2
$\lvert\alpha\rvert > 1$	No input applied

Table 2.1: Controller selection based on the value of α

$$Q = \mathrm{diag}\,(0.0001,\ 0.0001,\ 0.2,\ 3) \tag{2.9}$$
$$R = (10) \tag{2.10}$$
$$Q_2 = \mathrm{diag}\,(0,\ 0,\ 6,\ 0) \tag{2.11}$$
$$R_2 = (100) \tag{2.12}$$

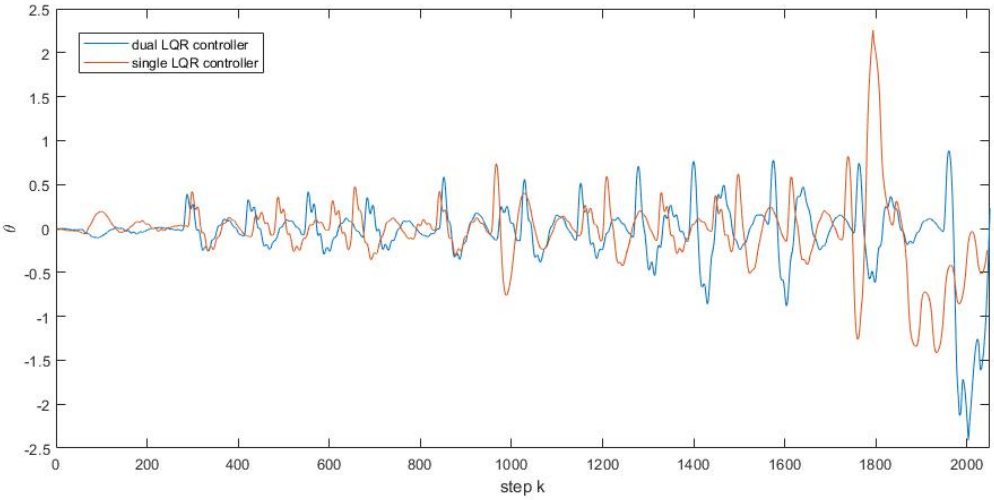

Figure 2.5: Single and dual LQR disturbance rejection

3

Model Predictive Control

3.1. MPC Definition

The base case MPC is the controller that acts optimally with respect to the optimization problem with prediction horizon N, system dynamics $x_{k+1} = f^{+}(x_k, u_k)$, input sequence $\bar{u}_N$ inequality constraints $g(\bar{u}_N)$, and equality constraints $h(\bar{u}_N)$ is defined as

$$\begin{aligned}
\mathrm{P_N}: \quad & \min_{\bar{u}_{N-1} \in \mathbb{U}} V_N(x_0, \bar{u}_{N-1}) && (3.1)\\
s.t. \quad & \phi_N(x_0, \bar{u}_{N-1}) \in \mathbb{X}_f && (3.2)\\
& g(\bar{u}_{N-1}) \leq 0 && (3.3)\\
& 0 = \bar{u}_{N-1} - \bar{K}_{N-1}\bar{x}_{N-1} && (3.4)\\
& && (3.5)
\end{aligned}$$

where the different terms are defined as the loss function, the evolution function of an LTI system and the terminal set. The, in general time variant, constraints are specific to the problem.

$$\phi_k(x_0, \bar{K}_{k-1}) = \left[\prod_{i=0}^{k-1} e^{Ah} - \left(e^{Ah} - I\right) A^{-1} B K_i\right] x_0 = \left[\prod_{i=0}^{k-1} F - G K_i\right] x_0 \tag{3.6}$$

which can be alternatively written for an input that is not expressed as state feedback as

$$\phi_k(x_0, \bar{u}_{k-1}) = F^k x_0 + \sum_{i=0}^{k-1} F^{K-1-i} G u_i \tag{3.7}$$

and the loss function is defined as

$$V_N = \bar{x}_{N-1}^{\top} Q_{N-1}^{e} \bar{x}_{N-1} + \bar{u}_{N-1}^{\top} R_{N-1}^{e} \bar{u}_{N-1} + x_N^{\top} P x_N \tag{3.8}$$

with

$$\bar{x}_{N-1} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ .. \\ x_{N-1} \end{pmatrix} = \begin{pmatrix} I \\ F - GK_0 \\ (F - GK_1)(F - GK_0) \\ .. \\ \prod_{i=0}^{N-2} F - GK_i \end{pmatrix} x_0 \tag{3.9}$$

$$\bar{Q}_{N-1} = I_{N-1} \otimes Q \tag{3.10}$$

$$\bar{u}_{N-1} = \begin{pmatrix} u_0 \\ u_1 \\ u_2 \\ .. \\ u_{N-1} \end{pmatrix} = -\bar{K}_{N-1}\bar{x}_{N-1} = -\begin{pmatrix} K_0 \\ K_1(F - GK_0) \\ K_2(F - GK_1)(F - GK_0) \\ .. \\ K_{N-1}\prod_{i=0}^{N-2}(F - GK_i) \end{pmatrix} x_0 \tag{3.11}$$

$$\bar{K}_{N-1} = diag\begin{pmatrix} K_0 & K_1 & .. & K_{N-1} \end{pmatrix} \tag{3.12}$$

$$\tag{3.13}$$

The set of admissible control actions is the set $\mathbb{U}$ and the state space is $\mathbb{X}$ Equation 3.14. The terminal set $\mathbb{X}_f$ is detailed in section 3.2

$$\mathbb{X} = \mathbb{X}_{\dot{\theta}} \times \mathbb{X}_{\dot{\alpha}} \times \mathbb{X}_{\theta} \times \mathbb{X}_{\alpha} \tag{3.14}$$

$$with \qquad \mathbb{X}_{\dot{\theta}=[-\dot{\theta}_{max},\dot{\theta}_{max}]} \tag{3.15}$$

$$\mathbb{X}_{\dot{\alpha}=[-\dot{\alpha}_{max},\dot{\alpha}_{max}]} \tag{3.16}$$

$$\mathbb{X}_{\theta} = [-\theta_{max}, \theta_{max}] \tag{3.17}$$

$$\mathbb{X}_{\alpha} = [-\alpha_{max}, \alpha_{max}] \tag{3.18}$$

3.2. Stability

Given the linearization is sufficiently accurate the stability of the unconstrained system is guaranteed to stabilize under the control low $\kappa : u = -Kx$ given the condition holds:

$$(A - BK_{N-1})^{\top}P(A - BK_{N-1}) - P + Q + K_{N-1}^{\top}RK_{N-1} \preceq 0 \tag{3.19}$$

with K being the solution to the LQR problem with an infinite horizon. For a finite horizon N, the stability under the control low $\kappa : u = -K_k x$ is guaranteed to be stable for the duration of the prediction, this up until N. Proceeding N the system's stability is guaranteed by proving that it enters the terminal set $\mathbb{X}_f$, where the terminal set is positive invariant with respect to the system $f^+(x, -K_{N-1}x)$ which means that all evolutions $\phi(x, -K_{N-1})$ for $x \in \mathbb{X}_f$ are contained within $\mathbb{X}_f$.

The system is linearized therefore the following condition needs to be also guaranteed for stability:

$$V_f(f^+(x, -K_{N-1}x)) \leq V_f(x) - l(x, -K_{N-1}x) \quad for\ all\ x \in \mathbb{X}_f \tag{3.20}$$

$$\tag{3.21}$$

In addition, a and P needs to be chosen such that the terminal set is contained within the state space and that the input is contained within the set of admissible input actions:

$$\mathbb{X}_f \in \mathbb{X} \tag{3.22}$$

$$-K_{N-1}\mathbb{X}_f \in \mathbb{U} \tag{3.23}$$

3.3. Reachable Set

Since the input is constrained and stability is guaranteed only if the evolution within the horizon results inside the terminal set, it is not possible for every initial state to do so. Therefore, in addition to the terminal set, the reachable set $\mathbb{X}_N$ is defined.

$$\exists\ \bar{u}_{N-1}\ \forall\ x \in \mathbb{X}_N \quad s.t.\ \phi_N(x, \bar{u}_{N-1}) \in \mathbb{X}_F \tag{3.24}$$

3.4. MPC Design

Given the model, the MPC definition and it's stability conditions the following parameters need to be designed:

- $\mathbb{X}$ the state space where the assumptions stand, assuming it is symmetric and that it is based on the one norm of the state there are 4 parameters: $\dot{\theta}_{limit}$, $\dot{\alpha}_{limit}$, θ_{limit}, α_{limit} to be determined
- u_{limit} the absolute value of the maximum and minimum input value
- h, the sampling time
- P, positive definite, which defines the terminal cost
- a which determines the terminal set together with P
- K_{N-1}, the last item of the control sequence that results in the transition to x_N
- L the observer gain, with $A - LC$ having poles at least 10 times faster than $A - BK_i$
- Q, positive definite, which defines the deviation from the origin of the state in the running cost
- R, positive definite, which defines the deviation from the origin of the input in the running cost

3.5. Off-line computations

Since MPC runs on-line the computational needs to be faster than the sampling. Thus certain parts of the MPC are computed first that are static in the process and therefore can be used repeatedly. First the terminal set is chosen to be characterized by the terminal cost V_f Equation 3.25 which provides an ellipsoid. In this region the value of the cost function is strictly decreasing which is used to bound the evolution of the states as a Lyapunov function.

$$a \geq x^\top P x \tag{3.25}$$

The terminal set needs satisfy the property of the Lyapunov decrease and the constraints

$$(A - BK_{N-1})^\top P(A - BK_{N-1}) - P + Q + K_{N-1}^\top R K_{N-1} \preceq 0 \tag{3.26}$$

The constrain $-K_{N-1}x \in \mathbb{U} \mid x \in \mathbb{X}_f$ is met by expressing boundary of $\mathbb{X}_f$ and finding the point on it that maximizes $-K_{N-1}x$ by finding the point on the hyperplane that is orthogonal to K_{N-1} by using the gradient which translates to

$$(-K_{N-1}x)_{max} \mid x \in \mathbb{X}_f = K_{N-1}\sqrt{a}\frac{aP_f^{-1}K_{N-1}^\top}{\sqrt{K_{N-1}P_f^{-1}K_{N-1}^\top}} \tag{3.27}$$

The same process is done to find the maximum value of each state within the terminal set, but since the states are linked to the construction to the problem it can be rewritten as:

$$(x_i)_{max} | x \in \mathbb{X}_f = \sqrt{aP_f^{-1}(i,i)} \tag{3.28}$$

where $P_f^{-1}(ii)$ stands for the i-th diagonal entry of the matrix P_f^{-1}. which results in the proper expression of the constraints.

The resulting $\mathbb{X}_f$ are presented for a 2D case for visualization Figure 3.1

Furthermore in order to use quadratic programming the problem is expressed as

$$\min u^\top H u \quad s.t.\ Fu \leq h \tag{3.29}$$

Where the Hessian matrix H is:

$$H = S_u^\top \bar{Q}_{N-1} S_u + \bar{R}_{N-1}$$

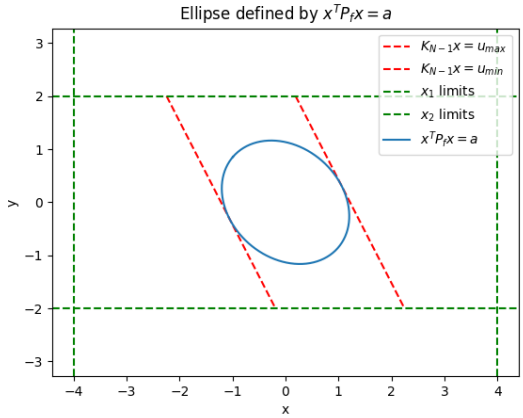

Figure 3.1: $\mathbb{X}_f$ 2D example

with S_u:

$$S_u = \begin{bmatrix} B & 0 & 0 & \cdots & 0 \\ AB & B & 0 & \cdots & 0 \\ A^2B & AB & B & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{N-1}B & A^{N-2}B & A^{N-3}B & \cdots & B \end{bmatrix} \in \mathbb{R}^{nN \times mN}$$

and with $\bar{Q}_{N-1}$:

$$\bar{Q}_{N-1} = \begin{bmatrix} Q & 0 & \cdots & 0 & 0 \\ 0 & Q & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & Q & 0 \\ 0 & 0 & \cdots & 0 & Q_f \end{bmatrix} \in \mathbb{R}^{nN \times nN}$$

and with $\bar{R}_{N-1}$:

$$\bar{R}_{N-1} = \begin{bmatrix} R & 0 & \cdots & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix} \in \mathbb{R}^{mN \times mN}$$

and F and h are given as

$$F = \begin{pmatrix} I_{N-1} \\ -I_{N-1} \\ S_u \\ -S_u \end{pmatrix} \tag{3.30}$$

$$h = I_{N-1} \otimes \begin{pmatrix} u_{limit} \\ -u_{limit} \\ x_{limit} \\ -x_{limit} \end{pmatrix} \tag{3.31}$$

3.6. Implementation

The following parameters are chosen for the MPC:

3.6.1. Constraints

The only constraint implemented is a bound on the input:

$$|u| \leq 0.5 \tag{3.32}$$

It is checked that constraints on the angles or velocities significantly degrade performance. The constraint that enforces that the terminal state is in the terminal set has not been implemented as it slows down the computation making it a quadratic constraint, quadratic programming problem. In some cases

it is necessary to implement the terminal constraint, but in the case of this project the terminal state was inside the terminal set without having to specify the constraint, therefore it is not present in the alogrithm.

3.6.2. Weight matrices

The best performing matrices are found to be the same as in LQR, with the terminal cost P from the solution of the Discrete Algebraic Riccati Equation.

$$Q = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad R = 100 \quad P = \begin{pmatrix} 2.4185 & 2.8477 & 18.7204 & 33.4939 \\ 2.8477 & 3.6088 & 21.3289 & 43.3278 \\ 18.7204 & 21.3289 & 178.3158 & 252.8892 \\ 33.4939 & 43.3278 & 252.8892 & 524.2353 \end{pmatrix} \tag{3.33}$$

The observer set-up is the same as in the LQR.

3.6.3. Horizon

Different horizon lengths are checked to determine the best one. A step response input of amplitude 0.7 is given to the controller to determine the best performance. The results can be seen in Figure 3.2.

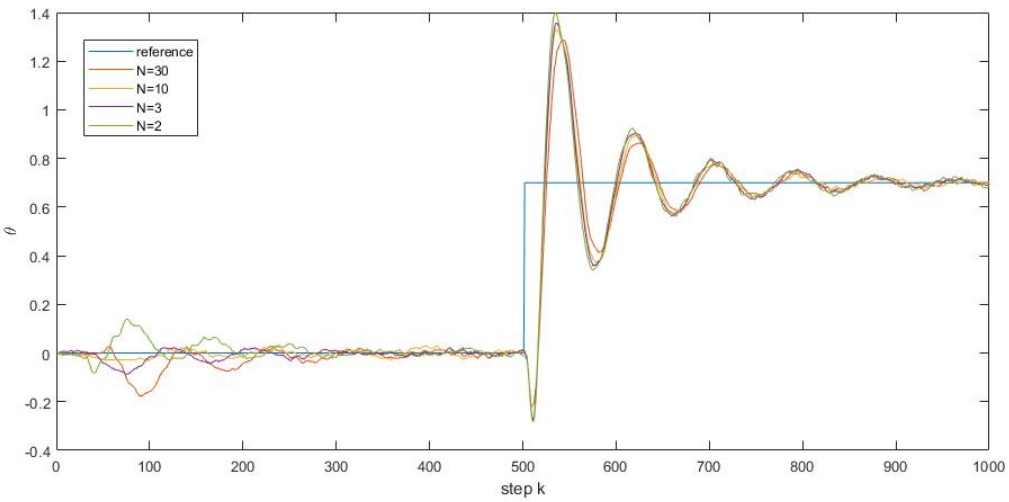

Figure 3.2: Step response of MPC for different N

It is found that for horizon lengths $N > 30$, the controller doesn't stabilise the system. Then $N = 30$ is the best performing controller. Then smaller horizons slightly increase the overshoot, not significantly though. The response is also very similar to that of the LQR, which makes sense. The only constraints implemented in MPC are the same as in LQR, makign the Riccati solution the optimal one.

4

Controller comparisons

The LQR and MPC controllers are now compared.

4.1. Sine input

First a zero input is given to the system for 4 seconds to allow it to stabilise around equilibrium. Then a sine with an amplitude of 0.7 and period of 2 seconds is given. Despite small oscillations, both controllers follow the reference well.

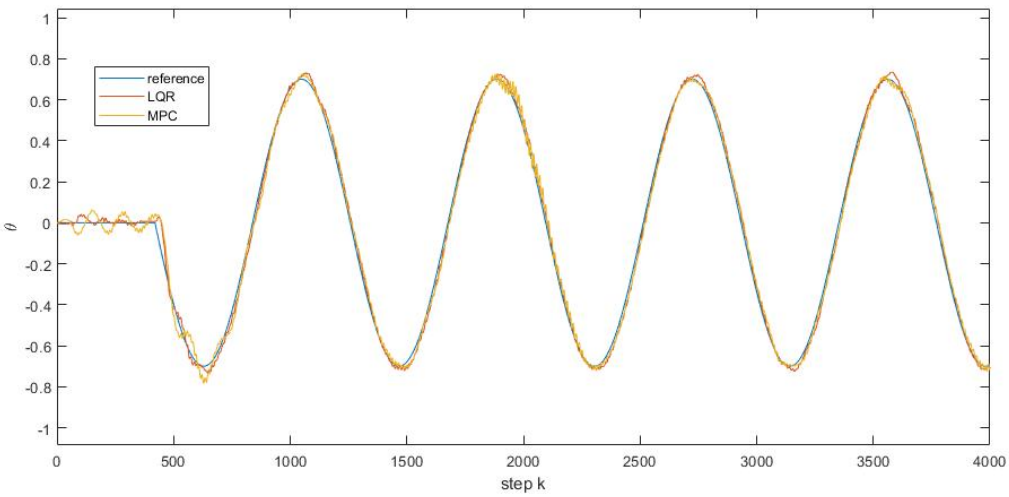

Figure 4.1: Responses of the LQR and MPC controllers to a sine input

4.2. Square input

Then a square wave is fed to the system, following of 4 seconds of settling. It is found that that the system does not stabilise for a pure square wave at the amplitude of 0.7. The system follows the first step well but then when a step in the opposite direction happens, system is too close to the end stops of the setup and too far from the equilibrium to go back ending in failure. The input is therefore decreased to the amplitude of 0.5 and a rate limiter block is added on the reference to ensure that the change in the input is not as sharp. Both controllers then follow the references relatively well.

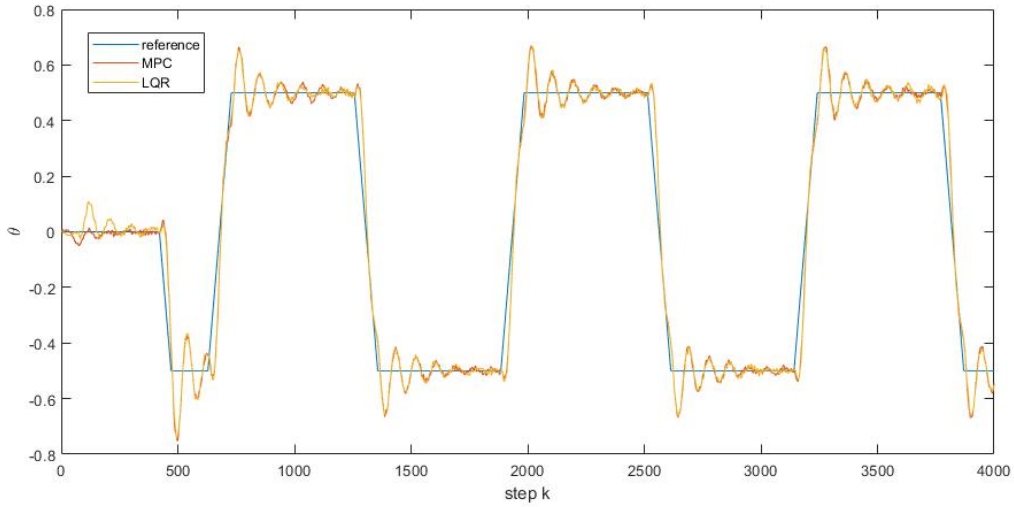

Figure 4.2: Responses of the LQR and MPC controllers to a square input